

5 The Impacts of AI, Robots, and Globalization on Labor Markets

Analysis of a Quantitative General Equilibrium Trade Model

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1 Introduction

A generally held view about the complementarity between labor and capital is that workers benefit from increased capital. However, the story can be very different when it comes to a special kind of capital, namely industrial robots, which the International Standard Organization (ISO) defines as ‘an automatically controlled, reprogrammable, multipurpose manipulator programmable in three or more axes, which can be either fixed in place or mobile for use in industrial automation applications’ (ISO 8373:2012). These industrial robots can perform tasks that factory workers would otherwise perform. Therefore, they may substitute for low-skilled, manual labor.

Technological progress has also occurred in areas that put high-skilled workers’ cognitive jobs at risk. In particular, artificial intelligence (AI) continues to advance at an amazing pace. It complements and substitutes for various types of workers in the medical and pharmaceutical industries among others, and changes future perspectives in many industries. Brynjolfsson and McAfee (2011) and Baldwin (2019) argue that robot technology and AI advance at an explosive pace so that only those that can utilize such new technology benefit. Meanwhile, the majority of workers may be left behind with lower pay.

Globalization, defined here as declining costs of trade in goods and services, has contributed to the global spread of robots and AI. World trade in robots has been steadily increasing since approximately 2000, as shown in Figure 5.1, although it still occupies a tiny portion (0.13% in 2018) of the world trade in capital goods. A notable feature of the robot and AI industries is that a small number of countries hold significant shares in world production and exports. Figure 5.2 shows the dominance of Japan and Germany in robot exports in 2018. AI companies are also geographically concentrated in a few countries such as the US, China, the UK, Germany, France, and India (Samoili et al., 2020).

Using a quantitative general-equilibrium trade model, this chapter assesses the impacts of robot/AI technological progress and globalization on labor markets in the world economy. In our model, robots can perform low-skilled labor’s tasks and AI high-skilled labor’s tasks in the manner of the

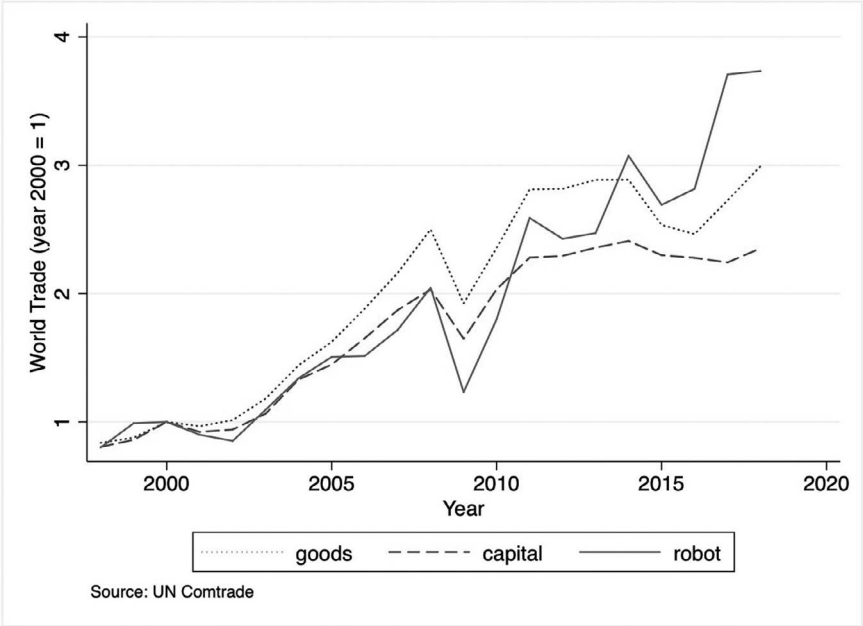


Figure 5.1 World Trade in Goods, Capital, and Robots

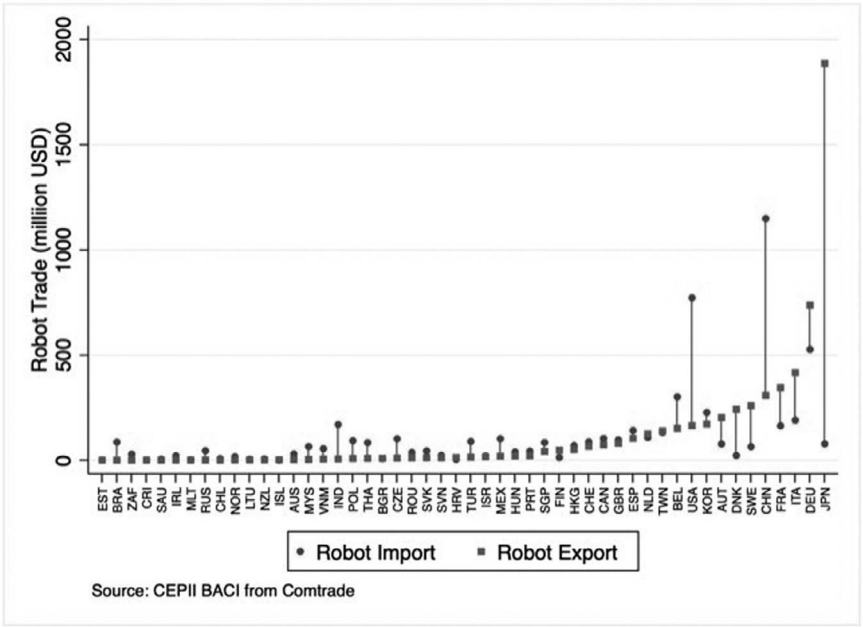


Figure 5.2 Robot Exports and Imports Across Countries in 2018

task-based approach developed by Acemoglu and Autor (2011), Acemoglu and Restrepo (2018a; 2018b), and Grossman and Rossi-Hansberg (2008, 2012). Following Eaton and Kortum (2002), Caliendo and Parro (2015), Sugita et al. (2019), and Furusawa and Sugita (2020), we also incorporate the important feature of global value chains (GVCs) linking multiple industries in multiple countries. A task-based model with a GVC structure enables us to address a rich set of questions regarding globalization, technology adoption, and income inequality.

In Section 2, we set up a model with robots but not AI and evaluate the effects of past technological progress and globalization. Section 3 presents our unique dataset that includes five production factors: two labor skill groups, robot installation, input-output tables, and bilateral trade for 17 ISIC 2-digit industries and 50 countries. We develop a novel method to use the data and the model's structure, notably the gravity equation of robot trade, in estimating two critical parameters: robot income shares in low-skilled task production in each industry and each country and elasticities of substitution between robots and low-skilled labor in each industry. The robot income shares are sizable in some industries but are generally small at the country level. The elasticities of substitutions are around a moderate number of 2. In Section 4, we first simulate the world economy in 2014, where either robot-related technology or trade costs are assigned their 1993 levels. The effect of advances in robot technology on labor markets is much smaller than that of trade liberalization from 1993 to 2014 despite being sizable in the most affected industries. Our results suggest that the macroeconomic impact of past robot technological progress on labor markets in 2014 was limited. Then, we simulate a future world where the task-productivity of robots becomes ten times its 2014 level. The share of tasks that robots would perform would still be small in most countries as would the impact on the labor market.

Finally, in Section 5, we extend our baseline model with robots to incorporate the substitution of AI for high-skilled labor. With limited data on trade in AI services, we estimate elasticities of substitution between AI and high-skilled labor in 17 industries and conduct a counterfactual analysis on the impact of a tenfold increase in AI task-productivity on labor markets. These elasticities of substitution are smaller than the estimated elasticities of substitution between robots and low-skilled labor. As a consequence, we find that its impact on the labor market is smaller than that of robotics and that advances in AI technology would decrease wage inequality between high-skilled and low-skilled labor.

The effects of automation on the labor market have been studied extensively. Autor et al.(2003) and Autor and Dorn (2013) show that automation replaces workers performing routine tasks, contributing to employment and wage polarization. Industrial robots primarily displace low-skilled production workers in factories. Graetz and Michaels (2018) and Acemoglu and Restrepo (2020) find that robot adoption reduces low-skilled workers' employment share and lowers employment and wages across commuting zones in the US, respectively.

However, they find evidence that robots raise firms' productivity (Acemoglu, Lelarge, and Restrepo, 2020b), which could benefit all workers, including the adversely affected low-skilled workers, by reducing consumer prices. Recent studies on AI also report similar job-replacing and productivity-enhancing effects.¹

Recent papers by Humlum (2019), Adachi (2021), and Artuc, Bastos, and Rijkers (2020) add new insights to the literature on the labor market impacts of robots in the world economy. Humlum (2019) constructs a dynamic general-equilibrium trade model that incorporates firms' choice of robot adoption and workers' choice of occupation and simulates the model with parameters estimated from Danish data to assess the impact of robot adoption on the Danish labor market. Adachi (2021) develops an open-economy model to analyze the impact of falling robot costs in Japan on the US labor market. He estimates elasticities of substitution between robots and labor across occupations and conducts a counterfactual analysis to assess the contribution of robots to wage polarization in the US. Like our model, Artuc et al. (2020) independently developed a quantitative Ricardian trade model with two-stage production that incorporates substitution of robots and labor. They analyze the effects of robotization in the North on trade patterns, wages, and welfare in the world.

Our analysis contributes to this new strand of the literature by enriching the model of the labor market and global trade structure to analyze the impact of globalization and the adoption of robots and AI on labor markets across the world. It is our contribution to estimate the robot shares of income at the country-industry-year level and use these estimates to estimate the elasticities of substitution between robots and low-skilled labor at the industry level. It is also worth conducting a rigorous model analysis with data and revealing the macroeconomic effect of robots and AI is small relative to the effect of globalization.

2 Model

We set up a model that is rich enough to quantify the effect of robotics and globalization on labor markets. The basic structure of our model is similar to those of Caliendo and Parro (2015), Sugita et al. (2019), and Furusawa and Sugita (2020). The fact that only a small number of countries export a large fraction of all robots traded in the world suggests that we need a multi-country model that incorporates differential industrial structures across countries. Building a model that allows differential industrial structures is also important because the use of robots (and AI) is heavily concentrated in some industries as we will see later. The model should also accommodate a GVC structure because its development is a prominent feature of globalization since the 1990s.

2.1 Basic Settings

N countries, indexed by i , $n \in \{1, \dots, N\}$, produce S goods and services, indexed by s , $k \in \{1, \dots, S\}$, for infinite time periods, indexed by $t \in \{0, \dots, \infty\}$. The

industry S represents the robot-producing industry. The model incorporates five factors of production: specific factor (G) only in the agriculture and mining industries, low-skilled labor (L), high-skilled labor (H), non-robot capital (K), and robots (R). Industry-specific factors and labor are supplied inelastically. Consumers invest and accumulate non-robot capital and robots. Goods are tradable across countries but services and factors are not. Robots and non-robot capital are tradable across countries only at the time of their investment. Once they are installed as capital goods, consumers can only lend them to domestic firms. All goods and factors are traded in perfectly competitive markets.

Each non-robot industry $s \in \{1, \dots, S-1\}$ produces two types of goods with different usages: final usage and intermediate usage. Final goods, denoted by f , are used only as final consumption and investment, whereas intermediate goods, denoted by m , are used only as inputs for production. Consumers do not consume robots. Goods of each usage $u \in \{f, m\}$ in industry s consist of a continuum of varieties $\omega^{su} \in [0, 1]$.

Country n 's representative consumer has a two-tier utility function with a Cobb-Douglas upper tier and a constant-elasticity-of-substitution (CES) lower tier:

$$U_n = \sum_{t=0}^{\infty} \beta^t \ln Q_{nt}^f, \quad Q_{nt}^f \equiv \prod_{s=1}^{S-1} (Q_{nt}^{sf})^{\alpha_{nt}^s},$$

$$Q_{nt}^{sf} \equiv \left[\int_0^1 q_{nt}^{sf}(\omega^{sf})^{\frac{\sigma^{sf}-1}{\sigma^{sf}}} d\omega^{sf} \right]^{\frac{\sigma^{sf}}{\sigma^{sf}-1}}, \quad (1)$$

where $\beta \in (0, 1)$ is the time discount factor, Q_{nt}^f the aggregate final consumption, Q_{nt}^{sf} the industry-level consumption and $\sum_{s=1}^{S-1} \alpha_{nt}^s = 1$. We let ω^{sf} denote a variety of final good s , $q_{nt}^{sf}(\omega^{sf})$ country n 's consumption of ω^{sf} at time t , and $\sigma^{sf} > 0$ the elasticity of substitution between varieties of final good s .

A firm in industry $s \in \{1, \dots, S\}$ in country n produces Υ_{nt} units of variety ω^{su} of usage u with the Cobb-Douglas production function:

$$\Upsilon_{nt} = A_{nt}^s Z_{sn}(\omega^{su}) G_{nst}^{\beta_{Gnt}^s} K_{nst}^{\beta_{Knt}^s} H_{nst}^{\beta_{Hnt}^s} T_{nst}^{\beta_{Tnt}^s} \prod_{k=1}^S (M_{nt}^{sk})^{\beta_{nt}^{sk}},$$

$$M_{nt}^{sk} \equiv \left[\int_0^1 m_{nt}^{sk}(\omega^{km})^{\frac{\sigma^{km}-1}{\sigma^{km}}} d\omega^{km} \right]^{\frac{\sigma^{km}}{\sigma^{km}-1}}, \quad (2)$$

where $\sum_v \beta_{vnt}^s + \sum_{k=1}^S \beta_{nt}^{sk} = 1$, A_{nt}^s is the country-industry specific component of the total factor productivity (TFP), $Z_{sn}(\omega^{su})$ the idiosyncratic component, and $\{G_{nst}, K_{nst}, H_{nst}\}$ the inputs of industry-specific factor, non-robot capital and high-skilled labor, respectively. Factor T_{nst} is a composite input of tasks that

either low-skilled labor or robots may perform, which we explain in detail in Section 2.2. M_{nt}^{sk} is a CES composite of intermediate good k where $m_{nt}^{sk}(\omega^{km})$ is the input of variety ω^{km} and $\sigma^{km} > 0$ the elasticity of substitution. The idiosyncratic productivity $Z_{sn}(\omega^{su})$ varies across varieties, independently following the Fréchet distribution $F^s(z) = \exp(-z^{-\theta^s})$ with industry-specific parameter θ^s .

Consumers and firms in country n purchase each variety ω^{su} with the lowest price that is given by $p_{nt}(\omega^{su}) \equiv \min_{i=1}^N p_{nit}(\omega^{su})$, where $p_{nit}(\omega^{su})$ is the unit cost of supply from country i to country n . There exists trade costs d_{nit}^{su} such that $p_{nit}(\omega^{su}) \equiv d_{nit}^{su} p_{iit}(\omega^{su})$. Trade costs consist of ad valorem tariffs τ_{nit}^s and iceberg non-tariff barriers D_{nit}^{su} such that $d_{nit}^{su} = (1 + \tau_{nit}^s) D_{nit}^{su}$. Trade costs are assumed to satisfy the triangle inequality $d_{njt}^{su} d_{jit}^{su} \geq d_{nit}^{su}$. Trade costs may differ between final usage and intermediate usage within industries.

2.2 Robot and Labor Substitution in Low-Skilled Task Production

We model the substitution of robots for low-skilled labor, following the task-based approach of Acemoglu and Restrepo (2018b; 2020). The task-based model can address that robots substitute labor for particular tasks, while simultaneously complementing workers for other tasks. The input of low-skilled tasks for the production of good s , T_{ist} in (2) is given by the Cobb-Douglas composite of a continuum of tasks $v \in [0, 1]$:

$$T_{nst} = \exp\left(\int_0^1 \ln T_{nst}(v) dv\right), \quad T_{nst}(v) \equiv \gamma_{st}(v) R_{nst}(v) + L_{nst}(v), \quad (3)$$

where $T_{nst}(v)$ is the amount of task v performed by either low-skilled labor or robots and $\gamma_{st}(v)$ represents the task-productivity of robots. Low-skilled labor and robots are perfect substitutes. Tasks are ordered by the comparative advantage of robots so that $\gamma'_{st}(v) < 0$ holds.

For a given low-skilled wage rate w_{Lnt} and robot rental rate w_{Rnt} , there exists a threshold task v_{nst} defined by $w_{Rnt}/\gamma_{st}(v_{nst}) = w_{Lnt}$. Robots perform tasks $v \leq v_{nst}$, while low-skilled workers perform $v > v_{nst}$. The threshold v_{nst} can be expressed as a function of the relative factor prices:

$$v_{nst} = v_{st}\left(\frac{w_{Rnt}}{w_{Lnt}}\right) \equiv \gamma_{st}^{-1}\left(\frac{w_{Rnt}}{w_{Lnt}}\right), \quad (4)$$

where γ_{st}^{-1} is the inverse function of γ_{st} , so that v_{st} is a decreasing function. From the optimal assignment of each task, the task composite (3) becomes

$$T_{nst} = \Gamma_{st}(v_{nst}) \left(\frac{R_{nst}}{v_{nst}}\right)^{v_{nst}} \left(\frac{L_{nst}}{1 - v_{nst}}\right)^{1 - v_{nst}}, \quad (5)$$

where $\Gamma_{st}(x) \equiv \exp\left(\int_0^x \ln \gamma_{st}(v) dv\right)$ captures the productivity advantage of robots over low-skilled labor.

The function v_{st} in (4) is the key function that determines the impact of robots on low-skilled labor's income. As seen in (5), the equilibrium value v_{nst} of function v_{st} determines the income shares of robots and low-skilled labor. In industry s in country n at time t , low-skilled workers receive the share $\beta_{Tnt}^s(1 - v_{nst})$ of the industry's revenue, while robots receive the share $\beta_{Tnt}^s v_{nst}$. We will estimate v_{st} in (4) from data.

2.3 Robot and Capital Accumulation

A representative consumer invests and accumulates robot capital and non-robot capital, lending them to local firms in their country. The robot stock R_{nt} and the non-robot capital stock K_{nt} in country n are accumulated as follows:

$$R_{nt+1} = I_{nt}^R + (1 - \delta_R)R_{nt} \text{ and } K_{nt+1} = I_{nt}^K + (1 - \delta_K)K_{nt}, \quad (6)$$

where I_{nt}^R and I_{nt}^K are investment in robot and non-robot capital, respectively, and δ_R and δ_K are the depreciation rates. For simplicity, the investment technology of non-robot capital is given by the same CES aggregator as the consumption aggregator Q_{nt}^f expressed by (1). Investment in robots is also expressed as a

CES aggregate of varieties of robots, $I_{nt}^R \equiv \left[\int_0^1 I_{nt}^R(\omega_R)^{\frac{\sigma_R-1}{\sigma_R}} d\omega_R \right]^{\frac{\sigma_R}{\sigma_R-1}}$, where $\sigma_R > 0$ is the elasticity of substitution and $I_{nt}^R(\omega_R)$ is the robot investment of variety ω_R .

A representative consumer's maximization problem can be expressed in two steps. Let w_{Knt} denote the rental rate of non-robot capita. Let P_{nt} and P_{nt}^R be the price index of consumption/capital investment and robot investment, respectively. The upper-tier problem is to choose the aggregate consumption Q_{nt} and investment (I_{nt}^R, I_{nt}^K) :

$$\begin{aligned} & \max_{\{Q_{nt}, I_{nt}^K, I_{nt}^R\}} \sum_{t=0}^{\infty} \beta_n^t \ln Q_{nt} \\ & \text{s.t. } P_{nt}(Q_{nt} + I_{nt}^K) + P_{nt}^R I_{nt}^R = w_{Knt} K_{nt} + w_{Rnt} R_{nt} + E_{nt} \text{ and (6),} \end{aligned}$$

where K_{n0} and R_{n0} are given, and E_{nt} represents income from other sources including labor income, distributed trade balance, and tariff revenue. We assume that the economy is in a steady state, where $K_{nt} = K_n$, $R_{nt} = R_n$, $Q_{nt} = Q_n$, $I_{nt}^K = \delta_K K_n$, and $I_{nt}^R = \delta_R R_n$. Then, the Euler equation gives us the following expressions for the rental rates of capital:

$$w_{Rnt} = P_{nt}^R(r_n + \delta_R) \text{ and } w_{Knt} = P_{nt}(r_n + \delta_K), \quad (7)$$

where $r_n \equiv (1 - \beta_n)/\beta_n$ is the real interest rate.

In the second-tier problem, the consumer chooses demands for individual varieties of goods, taking $\{Q_{nt}^s, I_{nt}^K, I_{nt}^R\}$ as given. It follows from Eaton and Kortum (2002) that the trade share of country i 's products of usage u in industry

s in country n is given by

$$\pi_{nit}^{su} = \frac{(A_{it}^s)^{\theta_s} (c_{it}^s d_{nit}^{su})^{-\theta^s}}{(P_{nt}^{su}/\gamma^{su})^{-\theta^s}}, \quad (8)$$

where γ^{su} is a constant, c_{it}^s is the value of the Cobb-Douglas function of factor prices and intermediate good price indexes (shown in (17) in the Appendix), and P_{nt}^{su} is the price index for goods of usage u in industry s in country i satisfying

$$\left(\frac{P_{nt}^{su}}{\gamma^{su}}\right)^{-\theta^s} = \sum_{i=1}^N (A_{it}^s)^{\theta_s} (c_{it}^s d_{nit}^{su})^{-\theta^s}. \quad (9)$$

The trade share equation (8) relates trade flows to prices in importing countries and leads to the gravity equation. We exploit this relationship to estimate robot prices from robot trade data.

2.4 Equilibrium Conditions for Counterfactual Analyses

Following Dekle et al. (2008) and Caliendo and Parro (2015), we consider a system of equilibrium conditions for changes in variables. We denote the counterfactual change in variable x by $\hat{x}_t \equiv x'_t/x_t$, where x_t is the value of variable x in equilibrium at time t while x'_t is its value in equilibrium of the counterfactual economy. We relegate to the Appendix the description of the system of equilibrium

Table 5.1 Countries in Our Dataset

Argentina	Hong Kong	Korea
Australia	Hungary	Romania
Austria	India	Russian Federation
Belgium	Indonesia	Singapore
Brazil	Ireland	Slovakia
Bulgaria	Israel	Slovenia
Canada	Italy	South Africa
Chile	Japan	Spain
China	Latvia	Sweden
Croatia	Lithuania	Switzerland
Czech Republic	Mexico	Taiwan
Denmark	Netherlands	Thailand
Estonia	New Zealand	Turkey
Finland	Norway	United Kingdom
France	Philippines	United States
Germany	Poland	Vietnam
Greece	Portugal	Rest of the world

conditions and relegate the conditions for changes in variables and our numerical-solution algorithm to the Online Appendix available on the authors' web pages.

3 Data and Estimation

3.1 Data

We construct a unique dataset for our quantitative general equilibrium analysis. The dataset includes information about bilateral trade in goods and robots, input-output tables, country-industry-year-level output and input allocations, and country-year-level factor prices for: 50 countries and the rest of the world (Table 5.1); 13 good industries and four service industries, which closely follow the 2-digit ISIC (International Standard Industrial Classification) codes; five production factors; and four years (2004, 2007, 2011, 2014).

This section explains the data on robots and labor skill groups. The data sources for other variables are conventional in the literature: the Global Trade Analysis Project (GTAP) database version 10, the Penn World Table, and the International Labour Organization (ILO) stat database. Details on the data construction are relegated to the Online Appendix.

Robot Stocks, Robot trade, and Robot Prices: the International Federation of Robotics (IFR) database reports the physical quantity of newly installed industrial robots for 71 countries and 18 industries since 1993. The IFR follows the ISO 8373 definition of robotics, mentioned in the Introduction, and has been used in the literature, e.g., Graetz and Michaels (2018) and Acemoglu and Restrepo (2020). Following the procedure in Graetz and Michaels (2018), we construct robot stocks using the perpetual inventory method with the depreciation rate $\delta_R = 0.1$. We collect bilateral trade in industrial robots from the UN Comtrade database, which contains the category of the 6-digit harmonized system code 847950, "industrial robots for multiple uses". We construct time-series data on the price for robots in Japan, unit robot price in Japan in 2014 calculated from the IFR data, and the time-series robot price indices with the benchmark of the 2015 price collected from the Bank of Japan (BoJ) Corporate Goods Price Index database. The BoJ monthly surveys prices of industrial robots and reports quality-adjusted price indices since 1990, which to our knowledge is the only data on quality-adjusted robot prices in our sample period.

Labor Skill Groups: following Weingarden and Tsigas (2010), we aggregate 10 occupations in the International Standard Classification of Occupations (ISCO-08) from the ILO database into two skill categories. The high-skill group includes: (1) Managers, (2) Professionals, and (3) Technicians and Associate Professionals. The low-skill group includes: (4) Clerical Support Worker, (5) Service and Sales Workers, (6) Skilled Agricultural, Forestry, and Fishery Workers, (7) Craft and Related Trades Workers, (8) Plant and Machine Operators and Assemblers, and (9) Elementary Occupations. We removed the category "Armed Forces Occupations" from the data.

3.2 Parameter Estimation

Cobb-Douglas Parameters: we calibrate the Cobb-Douglas parameters of utility and production functions $\{\alpha_{snt}, \beta_{snt}\}$ from expenditure shares calculated from the data.

Trade Elasticities We estimate Fréchet parameters θ^s , also called trade elasticities, by exploiting variations of bilateral tariffs in the gravity model, in the spirit of Caliendo and Parro (2015). Trade costs d_{nit}^{su} are modeled as $\ln d_{nit}^{su} = \ln(1 + \tau_{nit}^s) + \sum_k TC_{ni,k} \delta_{kt}^{su} + \tilde{\varepsilon}_{nit}^{su}$, where τ_{nit}^s is the bilateral tariff rate, $\tilde{\varepsilon}_{nit}^{su}$ the idiosyncratic trade costs, and $TC_{ni,k}$ the k -th variable of the country-pair characteristics commonly used in the gravity analysis.² Substituting $\ln d_{nit}^{su}$ into (8) yields the following gravity model with fixed effects:

$$\ln \pi_{nit}^{su} = \beta^s \ln(1 + \tau_{nit}^s) + \sum_t \sum_k TC_{ni,k} \left(\beta_{kt}^f + I_{\{u=m\}} \beta_{kt}^m \right) I_{\{year=t\}} + ex_{it}^s + im_{nt}^{su} + \varepsilon_{nit}^{su}, \quad (10)$$

where $I_{\{year=t\}}$ and $I_{\{u=m\}}$ are year dummies and intermediate goods dummies, respectively, ex_{it}^s the exporter-time fixed effect, and im_{nt}^{su} the importer-time-usage fixed effect.

We estimate (10) by OLS for each tradable-good industry (IFR industry from 1 to 13) separately, pooling all four periods (2004, 2007, 2011, 2014) and excluding observations without tariff information.³ Table 5.2 reports the estimated elasticities $\theta^s = -\beta^s$ with standard errors clustered at the exporter-importer-year level. The estimates have small standard errors and are generally higher (but within a comparable range) than those in other studies (e.g., Caliendo and Parro (2015)).⁴ For non-tradable service industries, we choose the median estimate of 10.582 as their θ^s , because our approach (as in other similar approaches) cannot be applied to non-tradable industries.

Table 5.2 Estimated Trade Elasticities

IFR	Description	θ	SE	<i>n.obs</i>
1	Agriculture, forestry, and fishing	4.456	(1.341)	15,940
2	Mining and quarrying	18.685	(5.029)	15,890
3	Food and beverages	8.429	(0.759)	15,952
4	Textiles	6.799	(0.721)	15,960
5	Wood and furniture	11.535	(1.663)	15,950
6	Paper	17.533	(1.743)	15,948
7	Plastic and chemical products	11.142	(1.087)	15,962
8	Glass, ceramics, stone, and mineral products	8.913	(1.172)	15,958
9	Metal	14.522	(1.459)	15,962
10	Electrical, electronics, and machinery	11.228	(1.212)	15,962
11	Automotive	10.582	(0.883)	15,950
12	Other vehicles	9.198	(1.621)	15,934
13	All other manufacturing branches	6.545	(1.017)	15,950

Notes: Standard errors (SE) are clustered at the exporter-importer-year level. All estimates are statistically significant at 1% level.

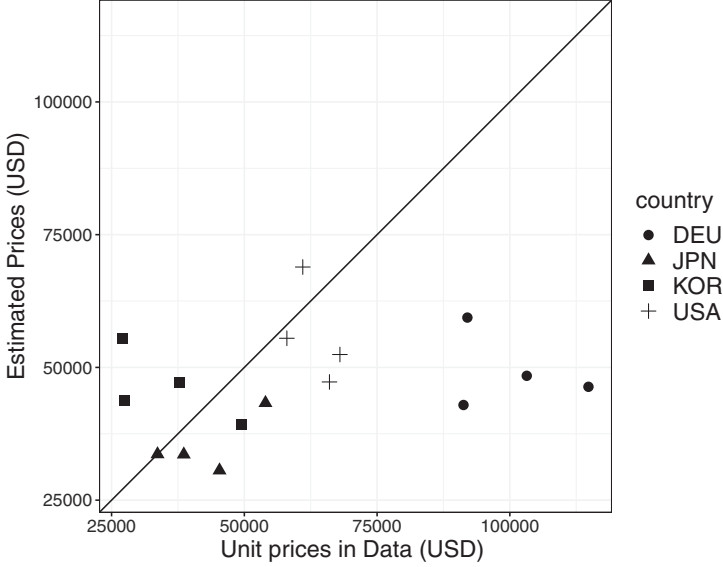


Figure 5.3 Estimated and Actual Robot Prices

Robot Prices and Production: we estimate the prices and production of robots from the gravity equation of trade in industrial robots. Let $X_{nt}^R \equiv X_{nnt}^R + \sum_{i \neq n} X_{nit}^R$ be country n 's total expenditure (i.e., investment) on robots. The trade share equation (8) implies the following gravity equation of robot trade:

$$\ln X_{nit}^R = \ln X_{nt}^R + \theta^R \ln P_{nt}^R + \theta^R \ln (A_{it}^R / c_{it}^R) - \theta^R \ln d_{nit}^R. \quad (11)$$

Modeling trade costs of robots as $\ln d_{nit}^R = \sum_k TC_{ni,k} \delta_{kt} + \tilde{\varepsilon}_{nit}$ as in (10), we estimate the gravity equation with fixed effects:

$$\ln X_{nit}^R = \zeta_{nt}^M + \zeta_{it}^E + \sum_k TC_{ni,k} \beta_{kt} + u_{nit}, \quad (12)$$

where ζ_{nt}^M is the importer-time fixed effect and ζ_{it}^E the exporter-time fixed effect.

We estimate (12) by the Poisson pseudo maximum likelihood (PPML), including the observations with zero trade. Choosing Japan as a benchmark country, we drop the constant term and the importer-year dummy ζ_{bt}^M for the country $b = JAPAN$ for each year.⁵ We see from (11) that the estimated coefficient of the importer-time dummy $\tilde{\zeta}_{nt}^M$ equals $\tilde{\zeta}_{nt}^M = \ln X_{nt}^R + \theta^R \ln P_{nt}^R - (\ln X_{bt}^R + \theta^R \ln P_{bt}^R)$.⁶ Rearranging this equation yields the price index of robots in country n relative to the benchmark country b :

$$\frac{P_{nt}^R}{P_{bt}^R} = \exp \left[\frac{1}{\theta^R} \left(\tilde{\zeta}_{nt}^M - \ln \frac{\sum_{i \neq n} X_{nit}^R + X_{nnt}^R}{\sum_{i \neq b} X_{bit}^R + X_{bnt}^R} \right) \right]. \quad (13)$$

Table 5.3 Robot Income Shares in Low-Skilled Tasks in 2014

<i>Country</i>	<i>Industry-Level Shares</i>			<i>Aggregate Share</i>
	<i>Median</i>	<i>Min</i>	<i>Max</i>	
	(1)	(2)	(3)	(4)
China	0.01%	0.00%	0.89%	0.04%
Germany	0.09%	0.00%	3.26%	0.21%
India	0.00%	0.00%	1.40%	0.02%
Indonesia	0.00%	0.00%	0.33%	0.02%
Japan	0.17%	0.00%	2.78%	0.29%
Korea	0.04%	0.00%	3.43%	0.31%
Singapore	0.19%	0.00%	5.20%	0.30%
Thailand	0.02%	0.00%	5.21%	0.24%
United States	0.01%	0.00%	1.13%	0.03%

We obtain robot prices and robot rental rates from (13). As an estimate of θ^R , we use the estimated trade elasticity in the IFR industry 10 “Electrical, electronics, and machinery” that includes industrial robots. We calculate domestic trade by $X_{nt}^R = \exp \left[\tilde{\zeta}_{nt}^E + \tilde{\zeta}_{nt}^M - \theta^R \ln d_{nt}^R \right]$ for each country n where we estimate domestic trade costs $\theta^R \ln d_{nt}^R$ by the gravity equation (12) for the IFR industry 10 using the domestic trade data.

Using the robot price in Japan P_{bt}^R , we can now calculate robot price P_{nt}^R for each country from (13). Figure 5.3 compares the estimated robot prices with the unit prices in the IFR database in the four countries, Germany, Japan, Korea, and the USA. The estimated robot prices are reasonably close to the actual ones, except in Germany.⁷ Finally, we calculate the robot rental rate w_{Rnt} from (7), where we collect the real interest rate r_{nt} from the World Development Indicators and choose a depreciation rate of $\delta^R = 0.1$, following Graetz and Michaels (2018).

Robot Income Shares: based on the robot-stock data and estimates of the robot rental rates, we calculate the robot income share in the income from low-skilled tasks, $v_{nst} = w_{Rnt} R_{nt}^s / (w_{Rnt} R_{nt}^s + w_{Lnt} L_{nt}^s)$ and report it for selected countries in 2014 in Table 5.3.⁸ Columns (1), (2), and (3) report the median, minimum, and maximum shares. Column (4) reports the aggregate robot income share, defined by $v_{nt} \equiv \sum_s w_{Rnt} R_{nt}^s / \sum_s (w_{Rnt} R_{nt}^s + w_{Lnt} L_{nt}^s)$. Table 5.3 illustrates that the income shares of industrial robots in low-skilled tasks are very small in most industries across all countries.⁹

3.3 Elasticities of Substitution between Robots and Low-Skilled Labor

We assume that the function v_{st} in (4) takes the following logistic function:

$$v_{nst} = v_{st} \left(\frac{w_{Rnt}}{w_{Lnt}} \right) = \frac{\exp \left(l_{st} - (\sigma_s - 1) \ln \frac{w_{Rnt}}{w_{Lnt}} + \epsilon_{nst} \right)}{1 + \exp \left(l_{st} - (\sigma_s - 1) \ln \frac{w_{Rnt}}{w_{Lnt}} + \epsilon_{nst} \right)}, \quad (14)$$

Table 5.4 Elasticities of Substitution Between Robots and Low-Skilled labor

IFR	Description	σ_s	Robust SE	1st Stage F	n.obs
1	Agriculture, forestry, and fishing	2.028	(0.209)	20.1	118
2	Mining and quarrying	1.435	(0.360)	15.4	82
3	Food and beverages	2.158	(0.173)	23.5	182
4	Textiles	2.879	(0.238)	14.9	120
5	Wood and furniture	2.927	(0.316)	21.5	140
6	Paper	1.947	(0.200)	17.9	133
7	Plastic and chemical products	1.775	(0.177)	25.2	189
8	Glass, ceramics, stone, and mineral products	1.995	(0.171)	24.5	170
9	Metal	2.181	(0.179)	25.0	189
10	Electrical, electronics, and machinery	2.065	(0.215)	20.8	176
11	Automotive	1.826	(0.203)	21.2	175
12	Other vehicles	1.793	(0.173)	28.8	170
13	All other manufacturing branches	1.906	(0.173)	22.8	178
14	All other non-manufacturing branches	1.221	(0.209)	24.8	132
15	Electricity, gas, and water supply	1.724	(0.152)	22.0	99
16	Construction	1.070	(0.195)	26.9	145
17	Education, research, and development	1.757	(0.202)	20.8	175

Notes: Standard errors are heteroscedasticity robust standard errors. The first stage F-values differ across industries because of the difference in the sample sizes.

where ι_{st} and ϵ_{nst} capture the sector-year specific determinants and the unobserved determinants of the robot demands other than the relative rental rate of robots, respectively.¹⁰ Given that the task production function (5) implies $v_{nst} = w_{Rnt} R_{nt}^s / (w_{Lnt} L_{nt}^s + w_{Rnt} R_{nt}^s) = (w_{Rnt} R_{nt}^s / w_{Lnt} L_{nt}^s) / [1 + (w_{Rnt} R_{nt}^s / w_{Lnt} L_{nt}^s)]$, formulating v_{st} by (14) is equivalent to formulating the relative demand function for robots to be log-linear:

$$\ln \frac{R_{nt}^s}{L_{nt}^s} = \iota_{st} - \sigma_s \ln \frac{w_{Rnt}}{w_{Lnt}} + \epsilon_{nst}, \quad (15)$$

where σ_s is the elasticity of substitution between robots and low-skilled labor. It follows from (14) that $\sigma_s > 1$ must hold for v_{st} to be a decreasing function.

Since ϵ_{nst} may be correlated with w_{Rnt} , we use an instrument (IV) for w_{Rnt} that shifts only the supply function. As such an IV, we use country n 's geographical access to robot exporters defined by $\zeta_{nt} \equiv \sum_{i \neq n} (A_{it}^R)^{\theta^R} (c_{it}^R d_{nit}^R)^{-\theta^R}$ and calculate $\tilde{\zeta}_{nt} = \sum_{i \neq n} X_{nit}^R / X_{nt}^R (P_{nt}^R)^{\theta^R}$ from the estimated robot gravity equation (12). Since the robot price satisfies $(P_{nt}^R / \gamma^R)^{-\theta^R} = \zeta_{nt} + (A_{it}^R)^{\theta^R} (c_{it}^R d_{nit}^R)^{-\theta^R}$ as shown in (9), it is clear that the IV ζ_{nt} utilizes the variation in the price of imported robots. Moreover, since the exporter-specific component $(A_{it}^R)^{\theta^R} (c_{it}^R)^{-\theta^R}$ of ζ_{nt} is common for all importers at time t , trade cost d_{nit}^R is the main source of cross-sectional variation of ζ_{nt} across countries. The exclusion restriction is that

trade cost d_{nit}^R is uncorrelated with ϵ_{nst} , the unobserved determinants of the robot demand.¹¹

We estimate (15) for each industry by the two-stage least squares with $\ln(\tilde{\zeta}_{nt}/w_{Lnt})$ an IV for $\ln(w_{Rnt}/w_{Lnt})$. The equation (15) includes only the year fixed effect, but not country fixed effects because the variation of d_{nit}^R in $\tilde{\zeta}_{nt}$ is mostly cross-sectional so that $\tilde{\zeta}_{nt}$ has little time-series variation. Table 5.4 reports estimates of σ_s with the heteroscedasticity-robust standard errors. The standard errors are small and the first stage F-statistics are all greater than 10. Point estimates of σ_s are greater than 1 in all industries.¹²

The elasticities of substitution in Table 5.4 can be compared with estimates in other studies. Adachi (2021) estimates elasticities of substitution between robots and labor within five occupation groups (production, transportation, other routines, services, abstract) in the US labor market. All of our estimates fall in the range of his estimates from 0.8 (abstract) to 4.29 (transportation). There are also studies that estimate the elasticities of substitution between capital and labor. Most of them report estimates that are smaller than 1 (e.g., Antras (2004)). Our estimates suggest that in many industries, robots and low-skilled labor are more substitutable than general capital and labor. This makes sense since robots generally can perform tasks more flexibly than other types of capital.

4 Counterfactual Analysis

We conduct counterfactual analyses to quantify the impact of past and future automation by robotics on the global economy. In Section 4.1, we simulate counterfactual world economies in 2014 where robot-related technology and costs of international trade are set at their levels in 1993. Whereas, in section 4.2, we simulate counterfactual world economies in 2014 where robot technology has advanced from its 2014 level.

4.1 The Impacts of Robotics and Globalization from 1993 to 2014

4.1.1 Changes in Trade Costs and Robot Technology from 1993 to 2014

To analyze the effect of progress in robot technology and the reduction in trade cost separately, we consider two counterfactual scenarios: only the robot technology is set at the 1993 level (*Robot effect*), and only trade costs are set at the 1993 level (*Trade effect*).¹³ As for the trade effect, we confine ourselves to examining the impact of a change in *international* trade costs relative to domestic trade costs.¹⁴

Trade Effect We estimate changes in trade costs between 1993 and 2014, following Novy (2013). Assuming that trade costs are bilaterally symmetric and common across usages (i.e., $d_{nit}^s \equiv d_{nit}^{sf} = d_{int}^{sf} = d_{nit}^{sm} = d_{int}^{sm}$) and that domestic trade costs remain the same (i.e., $d_{nn2004}^s = d_{nn1993}^s$), a change in trade costs

between 1993 and 2004 can be expressed from (8) as:

$$\hat{d}_{nit}^s = \frac{d_{ni1993}^s}{d_{ni2004}^s} = \left(\frac{X_{ii1993}^s X_{nn1993}^s X_{ni2004}^s X_{in2004}^s}{X_{ni1993}^s X_{in1993}^s X_{ii2004}^s X_{nn2004}^s} \right)^{1/(2\theta^s)}. \quad (16)$$

With the estimates of θ^s in Table 5.2, we estimate trade costs from the data on international and domestic trade: the CEPII TradeProd database (De Sousa, Mayer, and Zignago, 2012) and the International Trade and Production Database for Estimation (Borchert, Larch, Shikher, and Yotov, 2020).¹⁵

Robot Effect We calibrate the progress in robot-related technology from 1993 to 2014 by matching predicted values of two target variables in the model with the data. The first target is robot density, which is defined as the stock of industrial robots per 1000 workers. According to Graetz and Michaels (2018) and our own calculation from the IFR data, robot density in the world increased by 3.79-fold from 1993 to 2014. The second target is the price index of industrial robots in Japan, which declined from 100 in 1993 to 65.5 in 2014. The model predicts these changes, when the robot task-productivity and TFP for robot production were lower in 1993 than in 2014 by 35% and 31.5%, respectively, i.e., $\hat{\gamma}_{s2014}(v) = \lambda_{s2014} = 0.65$ and $\hat{A}_{n2014}^R = \hat{A}_{2014}^R = 0.685$. These counterfactual changes in robot technology are assumed to be common for all countries.

4.1.2 Results

This section reports the results of our counterfactual analyses about the effects on: (1) the relative price of robots to that of low-skilled labor and on robot density, (2) real wage rates for low-skilled and high-skilled workers, and (3) robot-worker replacement.

Table 5.5 The Impact of Robotics and Globalization from 2014 back to 1993: Robot Price and Robot Density

Country	Robot Rental/ Low-Skilled Wage		Robot Density (per 1000 workers)		
	Robot	Trade	2014	Robot	Trade
	(1)	(2)	(3)	(4)	(5)
China	+47.1%	+5.0%	0.221	-69.0%	-7.3%
Germany	+47.1%	+11.4%	4.793	-68.3%	-29.1%
India	+47.1%	+13.5%	0.022	-67.0%	-25.1%
Indonesia	+47.1%	-15.8%	0.039	-65.8%	+51.2%
Japan	+47.1%	+2.2%	8.386	-69.5%	-8.3%
Korea	+47.1%	+8.2%	6.331	-69.8%	-22.0%
Thailand	+47.1%	-21.0%	0.521	-66.3%	+10.7%
United States	+47.1%	-3.0%	1.525	-68.6%	+12.5%
World	+47.1%	+13.0%	0.740	-69.1%	+13.1%

Notes: Robot density is the number of industrial robots per 1000 workers (including both high-skilled and low-skilled workers). The value of “robot rental/low-skilled wage” in the world is the mean value of the countries in the sample.

Table 5.6 The Impact of Robotics and Globalization from 2014 back to 1993: Wages

Country	Real Wage for Low-Skilled		Real Wage for High-Skilled		Skill Wage Premium	
	Robot	Trade	Robot	Trade	Robot	Trade
	(1)	(2)	(3)	(4)	(5)	(6)
China	-0.01%	-4.96%	-0.03%	-5.37%	-0.02%	-0.43%
Germany	+0.03%	-5.89%	-0.07%	-5.00%	-0.10%	+0.95%
India	-0.00%	-6.37%	-0.01%	-2.37%	-0.01%	+4.28%
Indonesia	-0.01%	+0.64%	-0.02%	+0.64%	-0.01%	-0.00%
Japan	-0.01%	-2.14%	-0.11%	-1.50%	-0.10%	+0.66%
Korea	+0.00%	-4.59%	-0.16%	-4.88%	-0.17%	-0.30%
Thailand	+0.00%	+7.28%	-0.11%	+9.99%	-0.11%	+2.53%
United States	+0.00%	-0.47%	-0.02%	-0.74%	-0.02%	-0.27%
World	+0.01%	-4.15%	-0.04%	-2.92%	-0.04%	+1.16%

Notes: The values for the World are the mean values of the countries in the sample.

Columns (1) and (2) in Table 5.5 show the effect of robot technology and trade on the relative price of robots to that of low-skilled labor, w_{Rit}/w_{Lit} , for selected countries. Column (1) shows that the relative price of robots in 2014 would have been about 47% higher if robot technology was at the 1993 level. Although not shown in the table, our analysis reveals that the change in relative price is almost entirely attributed to the increase in robot rental rate. Column (2) indicates that the impact of trade costs are smaller than those of robot technology and vary across the countries. In most countries, the relative price of robots would have been higher if trade costs were at their 1993 levels. This is partly because a decline in trade costs lowered robot prices from 1993–2014. It is also because trade liberalization benefited low-skilled workers in those countries.

The impact on robot density reflects these changes in the relative price of robots, as illustrated in columns (4) and (5). Robot density would have been 69.1% lower on average in the world if robot technology was at its 1993 level. The impact of robot technology is similar in magnitude across countries, while that of trade costs is heterogeneous. The impact of robot technology and trade is rather large in most countries. Yet, the macroeconomic impact of increases in robot density can still be small, especially in countries with a small robot density in 2014 (see column (3)).

Let us turn to our main counterfactual analysis of the effects on labor markets. Table 5.6 shows the effects of robot technology and trade on the real wage rate for low-skilled labor, w_{Lit}/P_{it}^f , the real wage rate for high-skilled labor, w_{Hit}/P_{it}^f , and the skill wage premium, w_{Hit}/w_{Lit} . Robot installation directly displaces low-skilled labor. This is why the real wage rate for low-skilled labor would have been higher in Germany and the world average if robot technology was set at the 1993 level, though the general price level would have been higher in that case. The real wage rates for low-skilled labor would have been lower in some other countries, albeit only to a small degree. This suggests that robot installation increases

Table 5.7 The Impacts of Robotics and Globalization from 2014 back to 1993:
Robot-Worker Replacement

<i>Country</i>	<i>Industry Name</i>	<i>The most Robot Installing Industry</i>		
		<i>Robot</i>	<i>Change in Low-Skilled Labor</i>	
		<i>Change</i>	<i>Robot</i>	<i>Trade</i>
	(1)	(2)	(3)	(4)
China	Automotive	-50,933	65,934	1,752,039
Germany	Automotive	-85,275	11,362	-149,281
India	Automotive	-5,394	26,952	-138,858
Indonesia	Plastic and chemical products	-1,629	1,488	600,476
Japan	Electrical, electronics, and machinery	-130,283	10,572	-526,303
Korea	Electrical, electronics, and machinery	-68,512	8,872	-169,508
Thailand	Automotive	-6,972	17,055	-256,422
United States	Automotive	-73,910	7,147	127,833
World	Automotive	-488,763	213,092	504,859

Notes: The most robot-installing industry is an industry with the largest increase in robot stocks from the counterfactual equilibrium (where both robot and trade costs are set at their 1993 levels) to the 2014 equilibrium. The robot change is the change in the number of industrial robots in the most robot-installing industry. Columns (3) and (4) show the effects of changes in robot technology and trade costs on the number of low-skilled workers in the most robot-installing industry.

productivity, thereby benefiting all the workers regardless of their skill level. The effect of progress in robot technology on the real wage rate for high-skilled labor is positive in every country (as shown as negative numbers in Column (3)), reflecting the productivity effect and the complementarity between the low-skilled and high-skilled tasks. As a result, progress in robot technology increased the skill wage premium in every country in 1993–2014. This effect is relatively large in Japan and Korea, as the real wage rates for the high-skilled labor would have been much lower than in other countries if robot technology was at its 1993 level.

The effect of trade on real wage rates for both high-skilled and low-skilled labor is rather heterogeneous across countries, as shown in Columns (2) and (4). The effect of trade on the skill premium is also heterogeneous. The skill premium would have been lower in China, Korea, and the United States if trade costs were at their 1993 levels, which supports the argument that globalization entails wage inequality. But globalization appears to have mitigated wage inequality in other countries.¹⁶

The most notable result here, however, is that the impact of trade is generally much greater in size than robot technology. This is because robot shares are still small in 2014 (as shown in Table 5.3), so the impact on the labor market is limited.

Even though robotics has a minor impact on wages on the economy as a whole, it can have a substantial impact on some industries. Table 5.7 shows the industry that is estimated to have installed the largest number of robots

Table 5.8 Tenfold Increases in Robot Productivity: Robot Usage

<i>Country</i>	<i>Robot Task Share (v_{nst}) in the most Robot Installing Industry</i>			<i>Country-Level Robot Density (per 1000 Workers)</i>	
	<i>IFR</i>	<i>2014</i>	<i>CF</i>	<i>2014</i>	<i>CF</i>
	<i>(1)</i>	<i>(2)</i>	<i>(3)</i>	<i>(4)</i>	<i>(5)</i>
China	10	0.001	0.012	0.22	2.03
Germany	11	0.033	0.186	4.79	45.53
India	11	0.014	0.087	0.02	0.16
Indonesia	7	0.003	0.015	0.04	0.26
Japan	10	0.010	0.111	8.39	90.93
Korea	10	0.018	0.181	6.33	64.43
Thailand	11	0.052	0.273	0.52	3.68
United States	11	0.011	0.072	1.52	13.46
World	10	0.002	0.021	0.74	7.31

Notes: The most robot-installing industry is an industry with the largest increase in robot stocks from 2014 to the counterfactual equilibrium. Their IFR industry codes are shown in column (1). The values for the World are the mean values of the countries in the sample.

from 1993 to 2014 for each country and the change in the number of low-skilled workers employed in such industries. The automotive industry, plastic and chemical industries, and electrical, electronics, and machinery industries are most affected. The number of low-skilled workers in China, India, and Thailand among others would have been substantially higher in these respective industries if robot technology was at its 1993 level. It is worth emphasizing that in the automotive industry, which is one of the industries that install a large number of robots, robot technology displaced 213,092 low-skilled workers worldwide, while trade liberalization displaced 504,859 workers according to our simulation. In the automotive industry, the labor-displacement effect of robotics is rather large, nearly half the volume of that of trade.

4.2 *The Impacts of Future Robot Automation*

From the counterfactual analysis described in the previous section, we have found, among others, that progress in robot technology made only a minor impact on labor markets as a whole in the period of 1993–2014. This can be simply because the advances in robot technology were still at an early stage in 2014. In this section, therefore, we make a bold assumption that the robot productivity will increase tenfold (i.e., $\lambda_t = \hat{A}_t^S = 10$), from the 2014 level and assess the implications of progress in robot technology of that scale on the labor market.

Table 5.8 shows the change in the share of tasks performed by robots in the most affected industry for each country and the change in the country’s overall robot density from the 2014 economy to the counterfactual economy (CF). The worldwide robot task share in the electrical, electronics, and machinery industry would increase approximately tenfold from 0.002 to 0.021. It would

still be too small to make an impact on the macroeconomy, although the shares of the most affected industries would become rather sizable in some countries, such as Germany, Japan, and Thailand. The tenfold increase in robot technology would increase each country's robot share and robot density about tenfold, as columns (4) and (5) illustrate.

Such large changes in robot technology engender a much greater impact on the labor market than observed in the previous section's counterfactual analysis. Table 5.9 presents the aggregate labor market impact, showing the changes in the real wage rates for low-skilled and high-skilled labor and the skill premium. As robot technology advances, high-skilled workers would benefit in all countries. Low-skilled workers would also benefit in all countries in this list. The skill premium would increase in all countries in the list but India. The tenfold increase in robot technology would make sizable positive impacts on the real wage rate for high-skilled labor and hence increase the skill premium by non-negligible percentage points in countries such as Germany, Japan, and Korea. But, as a whole, even the tenfold increase in robot productivity would make only a moderate impact on the global labor market.

We also investigate to what degree low-skilled workers would be relocated as a result of the technological progress, which is reported in column (4). In some countries, such as Germany, Japan, and Korea, a sizable percentage of workers would be relocated.¹⁷ But, we see that the impact of progress in robot technology on the labor market would still not be that large on average.

5 AI

This section extends our analysis of robots to AI. We assume here that AI can perform the high-skilled labor's tasks H_{Hst} and reformulate the production function (3) in the same manner as robots substitute for the low-skilled workers.

Unlike robots that are a special sort of capital, we treat AI as a service to which user firms subscribe. Thus, we directly calculate each country's expenditure on AI to derive the share of AI in high-skilled tasks. The difficulty, though, is that we do not have data on such expenditures, specifically on AI. AI trades are recorded in the GTAP database as part of trades in the communication sector, which includes postal services, motion pictures, and others.

To find an appropriate share of AI in the communication sector, we make a bold assumption that the world average share of AI in the most AI installing Industry, namely "All other non-manufacturing branches" (IFR industry 14), equals 0.002, the world average share of robots in the most robot-installing industry. Using the World Input-Output Database (WIOD) Release 2016, we calculate the trade shares of "computer programming, consultancy, and the related activities; information service activities" in those in the communication sector. When only 1.5% of those shares are related to AI, the world average share of AI in the most AI installing Industry equals 0.002.

We estimate the elasticities of substitution between AI and the high-skilled labor in 17 IFR industries, in the same manner as we have done with robots.

Table 5.9 Tenfold Increases in Robot Productivity: Labor Market Impacts

	<i>Real Wage for Low-Skilled</i>	<i>Real Wage for High-Skilled</i>	<i>Skill Wage Premium</i>	<i>Aggregate Low-Skilled Labor Relocation</i>
<i>Country</i>	(1)	(2)	(3)	(4)
China	+0.55%	+0.90%	+0.35%	0.32%
Germany	+0.60%	+2.04%	+1.44%	0.94%
India	+0.17%	+0.07%	-0.10%	0.19%
Indonesia	+0.30%	+0.47%	+0.17%	0.30%
Japan	+2.12%	+3.30%	+1.16%	1.01%
Korea	+1.74%	+4.41%	+2.62%	1.38%
Thailand	+1.22%	+2.22%	+0.99%	0.59%
United States	+0.19%	+0.47%	+0.28%	0.20%
World	+0.39%	+0.99%	+0.60%	0.54%

Notes: The skill wage premium is the ratio of the high-skilled wage rate to low-skilled wage rate. The aggregate low-skilled labor relocation is calculated by $\sum_{s=1}^S L_{i2014}^s - L_{i2014}^s / 2L_{i2014}^s$ and shown as a percentage, where L_{i2014}^s and L_{i2014}^s denote the actual and counterfactual numbers of low-skilled workers employed in industry s in 2014, respectively. We divide the sum of changes in employment over the industries by 2 to avoid double counting. The values for the World are the mean values of the countries in the sample.

Table 5.10 shows the results. In every industry, the estimated elasticity is smaller than that of robots and low-skilled labor, suggesting that the high-skilled labor is less substitutable by AI than the low-skilled labor is by robots. Indeed, the estimated elasticities are smaller than 1 in ten industries. In the following simulations of the 2014 economy with enhanced AI productivities, we replace those

Table 5.10 Elasticities of Substitution Between AI and High-Skilled Labor

<i>IFR</i>	<i>Description</i>	σ_{AL}	<i>Robust SE</i>	<i>1st Stage F</i>	<i>n.obs</i>
1	Agriculture, forestry, and fishing	0.958	(0.087)	162.61	204
2	Mining and quarrying	1.113	(0.093)	162.61	204
3	Food and beverages	1.026	(0.071)	162.61	204
4	Textiles	0.936	(0.077)	162.61	204
5	Wood and furniture	0.873	(0.078)	162.61	204
6	Paper	1.033	(0.076)	162.61	204
7	Plastic and chemical products	1.045	(0.076)	162.61	204
8	Glass, ceramics, stone, and mineral products	0.824	(0.073)	162.61	204
9	Metal	0.710	(0.071)	162.61	204
10	Electrical, electronics, and machinery	0.727	(0.076)	162.61	204
11	Automotive	0.506	(0.089)	162.61	204
12	Other vehicles	0.740	(0.0923)	162.61	204
13	All other manufacturing branches	0.828	(0.073)	162.61	204
14	All other non-manufacturing branches	1.192	(0.072)	162.61	204
15	Electricity, gas, and water supply	1.189	(0.088)	162.61	204
16	Construction	0.956	(0.091)	162.61	204
17	Education, research, and development	1.251	(0.080)	162.61	204

Notes: Standard errors are heteroscedasticity robust standard error. The first stage F-values are identical for all industries because the first stage is common for all industries.

Table 5.11 Tenfold Increases in AI Productivity: AI Shares

<i>Country</i>	<i>AI Task Share in the most AI Subscribing Industry</i>		
	<i>IFR</i>	<i>2014</i>	<i>CF</i>
	(1)	(2)	(3)
China	14	0.0007	0.0011
Germany	14	0.0031	0.0048
India	17	0.0003	0.0005
Indonesia	14	0.0006	0.0009
Japan	14	0.0020	0.0032
Korea	14	0.0011	0.0017
Thailand	14	0.0004	0.0006
United States	14	0.0007	0.0011
World	14	0.0020	0.0030

Notes: The values for the World are the mean values of the countries in the sample.

elasticities with 1 because our model requires the elasticity to be greater than or equal to 1.¹⁸

With these estimates of the elasticities of substitutions, we follow the same procedure conducted in the analysis of robots and obtain the estimated impact on labor markets. To isolate the effect of AI on labor markets, we simulate the 2014 economy with the AI productivities (AI's counterpart of γ_{st}) that is 10 times as large as those in 2014, but leave the robot productivities at 2014 levels. Table 5.11 shows simulated changes in AI task shares. Compared with Table 5.8, resulting changes in the AI shares are much smaller than those in the robot shares when the robot productivity becomes tenfold, reflecting the observation that the elasticity of substitution between AI and high-skilled labor is smaller than that between robots and low-skilled labor in every industry.

Table 5.12 shows the estimated impacts of a tenfold increase in AI productivity on labor markets. As expected, low-skilled workers benefit from the progress in AI technology. The impact on real wage rates for low-skilled labor, however,

Table 5.12 Tenfold Increases in AI Productivity: Labor Market Impacts

<i>Country</i>	<i>Real Wage for Low-Skilled</i>	<i>Real Wage for High-Skilled</i>	<i>Skill Wage Premium</i>
	(1)	(2)	(3)
China	+0.07%	-0.04%	-0.11%
Germany	+0.56%	+0.45%	-0.11%
India	+0.07%	-0.04%	-0.11%
Indonesia	+0.07%	-0.08%	-0.15%
Japan	+0.37%	+0.26%	-0.11%
Korea	+0.14%	+0.02%	-0.12%
Thailand	+0.02%	-0.08%	-0.11%
United States	+0.18%	+0.07%	-0.11%
World	+0.35%	+0.20%	-0.14%

Notes: The values for the World are the mean values of the countries in the sample.

are smaller than those of robots on the real wage rate for high-skilled labor, as shown in Table 5.9, due to the smaller elasticities of substitution between AI and high-skilled labor. The impacts are relatively large in Germany and Japan, both of which have relatively large AI task shares both in 2014 and in the counterfactual equilibrium, as shown in Table 5.11. Similarly to the case of robotics, progress in AI productivity increases the real wage rates for high-skilled labor, which AI substitutes, in some countries such as Germany and Japan. Wage inequality would be reduced in all the countries.

6 Conclusion

This chapter has quantified the impact of past and future robot/AI progress on labor markets in a general equilibrium trade model with task substitutions between robots and low-skilled labor and between AI and high-skilled labor. Based on our new estimates of robot/AI income shares and elasticities of substitution for labor, the model shows that past technological progress of industrial robots had much smaller macroeconomic impacts on labor markets than the past trade liberalization and that the future impact of tenfold increases in the productivity of robot and AI on would be modest.

As Brynjolfsson and McAfee (2011) emphasize, robot technology advances at an exponential pace. It is likely that the benchmark year of 2014 was still on the verge of rapid technological advances, so our simulation may well have underestimated the future impact of robots and AI. The narrow definition of robotics may also have contributed to the underestimation. Technological progress in AI and robots have indeed been more rapid than we had anticipated. These are likely to reinforce each other in the technology-development stage and also at the stage of industrial usage. As Brynjolfsson and McAfee (2011) and Baldwin (2019) argue, the impacts of AI and robotics on the labor market, and more broadly the global economy, may soon become quite large.

Notes

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- 1 Acemoglu et al. (2020a) show that establishments whose workers engage in tasks compatible with current AI capabilities increase the hiring of workers in AI positions but decrease non-AI hiring. They also observe no discernible industry-level impact on employment or wages. Frey and Osborne (2017) assess how 702 occupations are susceptible to computerization and find evidence that wages and educational attainment exhibit a negative relationship with the susceptibility of computerization. Felten et al. (2018) assess which occupations are heavily affected by AI advances.

- 2 TC_{ni} includes the logarithm of distance, dummy variables for contiguity, common language, ever-colonial relationship, and a dummy indicating international trade (i.e., $i \neq n$) as opposed to intranational trade. These variables are from the CEPII datasets.
- 3 We also estimated (10) by the Poisson pseudo maximum likelihood (PPML) and found the PPML estimates slightly greater than those in Table 5.2.
- 4 Our estimates of trade elasticities are robust to the removal of observations with small trade shares. The Online Appendix shows the comparison of our estimates with those of some other studies: Caliendo and Parro (2015), Giri et al. (2021), and Shapiro (2016). It also shows the results of the same counterfactual analysis as presented in Section 4.1 but with mean estimates by those studies instead of our own. With smaller trade elasticities, price variation would be greater, and hence the impact of robotics and trade on real wage rates would tend to be greater. But, our main message about the size of the macroeconomic impact of robotics would remain valid.
- 5 If we included the constant term in (12), we would have to drop one importer-year dummy and one exporter-year dummy for each year to avoid perfect collinearity. Instead, we choose to drop the constant term and the benchmark country's importer-time dummy.
- 6 From the comparison of the gravity equation (12) with its theoretical counterpart (11), especially when $n = b$, it is apparent that the coefficient of the exporter-time dummy equals $\tilde{\zeta}_{it}^E = \theta^R \ln(A_{it}^R/c_{it}^R) + (\ln X_{bt}^R + \theta^R \ln P_{bt}^R)$. By including the exporter-time dummy for all exporting countries without the constant term (which would reflect the benchmark country's importer-time fixed effect), the coefficient of the exporter-time dummy reflects the benchmark country's importer-time fixed effect as well as the exporter-time fixed effect in question.
- 7 The prices in Germany are underestimated, possibly because German robots are of higher quality. If we remove the data for Germany, the root mean squared errors would be 13,806 dollars, which is slightly lower than one standard deviation of 14,439 dollars in the sample. The correlation between the estimated prices and the prices in the data would be 0.39.
- 8 The results of this estimation and all subsequent analyses for other countries are reported in the Online Appendix.
- 9 The estimated income shares of robots are so small partly because the IFR's definition of industrial robots is very narrow. The definition excludes machines that only work in specialized tasks.
- 10 A limitation of the logistic formulation (14) is that it always predicts a strictly positive $v_{nst} > 0$, even though the data includes $v_{nst} = 0$ for some industries in some countries. In our counterfactual analysis, we deal with this zero robot-use case in the two ways. In the first approach, we set $v_{nst} = 0$ and $\hat{v}_{nst} = 1$, whenever $v_{nst} = 0$. In the second approach, we replace $v_{nst} = 0$ with $\min\{v_{ist} > 0 : i = 1, \dots, N\}$. We take the first approach in a simulation where robot usage is expected to decrease in the counterfactual scenario and take the second approach otherwise.
- 11 The exclusion restriction would be violated if, for example, the robot productivity is heterogenous across countries and correlated with the distance to robot exporting countries.
- 12 They are statistically greater than 1 at 1% level of significance in all but three industries, 2 (Mining and quarrying), 14 (All other non-manufacturing branches), and 16 (Construction).
- 13 We also examined another scenario to show the *Combined effect*, where both robot technology and trade costs are set at their 1993 levels. The combined effect is not very different from the sum of the two individual effects, suggesting the interaction of the two effects is not large in scale.

- 14 It is difficult to estimate changes in domestic and international trade costs separately with our data, because the trade share of a country's products in a market, which we use to estimate the elasticities of substitution between input varieties in production, depends on the relative costs of international trade to those of domestic trade but not on them separately.
- 15 We connect the two datasets in the year 2000. Equation (16) cannot be applied if either X_{nit}^s or X_{int}^s is zero or missing. In such cases, we choose the median trade costs for other products between the same pair of countries.
- 16 Our model allows countries to have different factor endowments as well as different production technologies. Therefore, part of the heterogeneity of this impact of trade across countries can be explained by the traditional Stolper-Samuelson effect. The effect of trade liberalization on factor prices is complex (see, for example, Grossman and Rossi-Hansberg (2008); and Feenstra (2015), Chapter 4). It is beyond the scope of the chapter to identify the main cause of each specific result as to the effect of trade on real wage rates.
- 17 In terms of the number of workers, 465,418 workers in Japan and 165,852 workers in the United States, for example, would be relocated.
- 18 As shown in (14), if the elasticity of substitution is smaller than 1, a decrease in the relative price of AI would decrease the share of tasks that are performed by AI in our model with a Cobb-Douglas production of the composite task as formulated in (3). We imposed the restriction that the elasticity is greater than 1 to prevent such implausible demand responses to the AI price shocks from happening in our simulations.
- 19 We introduce O_{nt} to match country n 's total expenditure with its total income in each year t .

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A APPENDIX: EQUILIBRIUM CONDITIONS

From the production functions (2), the unit cost index c_{it}^s in (8) is given by

$$c_{it}^s = \zeta_i^s w_{Git}^{\beta_{Git}^s} w_{Hit}^{\beta_{Hit}^s} w_{Kit}^{\beta_{Kit}^s} w_{Tist}^{\beta_{Tist}^s} \prod_{k=1}^S (P_{it}^{km})^{\beta_{it}^{sk}}, \quad (17)$$

where ζ_i^s is a constant, w_{Git} the industry-specific factor price and w_{Hit} the high-skilled labor wage. From (5), the unit cost of the set of low-skilled tasks w_{Tist} in industry s in country i at time t becomes

$$w_{Tist} = \frac{(w_{Rit})^{v_{ist}} (w_{Lit})^{1-v_{ist}}}{\Gamma_{st}(v_{ist})}. \quad (18)$$

Let X_{nt}^{su} be country n 's tariff-inclusive expenditure on usage u in industry s . Let Υ_{nt}^s be the tariff-exclusive gross revenue of industry s in country n that satisfies $\Upsilon_{nt}^k = \sum_{i=1}^N \frac{\pi_{int}^{kf}}{1+\tau_{int}^k} X_{it}^{kf} + \sum_{i=1}^N \frac{\pi_{int}^{km}}{1+\tau_{int}^k} X_{it}^{km}$. The Cobb-Douglas production and utility functions imply

$$\begin{aligned} X_{nt}^{sm} &= \sum_{k=1}^S \beta_{nt}^{ks} \Upsilon_{nt}^k, \\ X_{nt}^{sf} &= \alpha_n^s (V_{nt} + TR_{nt} + TD_{nt} - X_{nt}^s) \text{ for } s = 1, \dots, S-1, \\ X_{nt}^{sf} &= I_{nt}^R = \delta^R R_{nt} \text{ and } X_{nt}^{sm} = 0, \end{aligned} \quad (19)$$

where $V_{nt} \equiv \sum_s w_{Hst} G_{nst} + w_{Hnt} H_{nt} + w_{Lnt} L_{nt} + w_{Knt} K_{nt} + w_{Rnt} R_{nt}$ is the factor income, $TR_{nt} \equiv \sum_{s=1}^S \sum_{i=1}^N \frac{\tau_{nit}^s}{1+\tau_{nit}^s} (\pi_{nit}^{sf} X_{nt}^{sf} + \pi_{nit}^{sm} X_{nt}^{sm})$ the sum of the tariff revenue, O_{nt} other exogenous income sources, and $TD_{nt} = \sum_{s=1}^S \sum_{i=1}^N \left(\frac{\pi_{nit}^{sf} X_{nt}^{sf} + \pi_{nit}^{sm} X_{nt}^{sm}}{1+\tau_{nit}^s} - \frac{\pi_{int}^{sf} X_{it}^{sf} + \pi_{int}^{sm} X_{it}^{sm}}{1+\tau_{int}^s} \right)$ the trade deficit exogenously given.¹⁹

For the production factors, the Cobb-Douglas production function implies

$$\begin{aligned} \beta_{Gnt}^s \Upsilon_{nt}^s &= w_{Gnt} G_{nst}, \\ \sum_{s=1}^S \beta_{Hnt}^s \Upsilon_{nt}^s &= w_{Hnt} H_{nt}, \quad \sum_{s=1}^S \beta_{Knt}^s \Upsilon_{nt}^s = w_{Knt} K_{nt}, \\ \sum_{s=1}^S \left(1 - v_{st} \left(\frac{w_{Rnt}}{w_{Lnt}} \right) \right) \beta_{Tnt}^s \Upsilon_{nt}^s &= w_{Lnt} L_{nt}, \text{ and } \sum_{s=1}^S v_{st} \left(\frac{w_{Rnt}}{w_{Lnt}} \right) \beta_{Tnt}^s \Upsilon_{nt}^s = w_{Rnt} R_{nt}. \end{aligned} \quad (20)$$

Since the consumer price index is $P_{nt} = \prod_{s=1}^{S-1} (P_{nt}^{sf})^{\alpha_s}$, the Euler equations (7) are written as:

$$\prod_{s=1}^{S-1} (P_{nt}^{sf})^{\alpha_s} = \frac{w_{Knt}}{r_n + \delta_K} \text{ and } P_{nt}^R = \frac{w_{Rnt}}{r_n + \delta_R}. \quad (21)$$

Conditions (9), (8), (17), (19), (20) and (21) determine the equilibrium allocation.